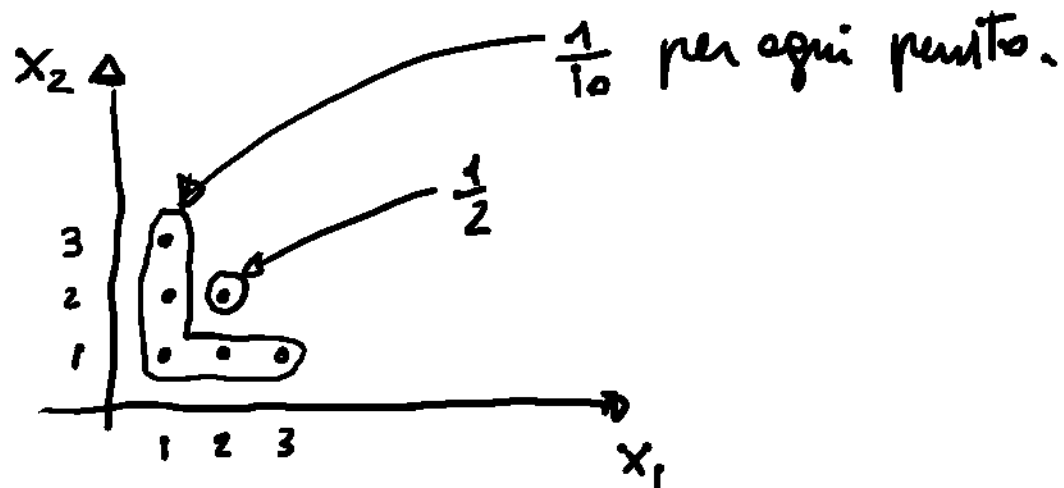


ESERCIZIO

Consideriamo la seguente densità congiunta



Calcolare $E[X_1 + X_2]$

SVOLGIMENTO

Y assume valori in $\{2, 3, 4\}$

$$P_Y(2) = P_X(1,1) = \frac{1}{10} \quad \text{SOMMA} = 1 \quad \text{OK}$$

$$P_Y(3) = P_X(1,2) + P_X(2,1) = \frac{1}{10} + \frac{1}{10} = \frac{2}{10}$$

$$P_Y(4) = P_X(1,3) + P_X(2,2) + P_X(3,1) = \frac{1}{10} + \frac{1}{2} + \frac{1}{10} = \dots = \frac{7}{10}$$

$$\begin{aligned} E[Y] &= 2 \cdot P_Y(2) + 3 \cdot P_Y(3) + 4 \cdot P_Y(4) = 2 \cdot \frac{1}{10} + 3 \cdot \frac{2}{10} + 4 \cdot \frac{7}{10} = \\ &= \frac{2 + 6 + 28}{10} = \frac{36}{10} = \frac{18}{5} \end{aligned}$$

METODO ALTERNATIVO

$$\begin{aligned} E[Y] &= E[X_1 + X_2] = (1+1) p_{\underline{X}}(1,1) + (1+2) p_{\underline{X}}(1,2) + (2+1) p_{\underline{X}}(2,1) + \\ &\quad + (3+1) p_{\underline{X}}(3,1) + (2+2) p_{\underline{X}}(2,2) + (1+3) p_{\underline{X}}(1,3) = \\ &= 2 \cdot \frac{1}{10} + 3 \cdot \frac{1}{10} + 3 \cdot \frac{1}{10} + \\ &\quad + 4 \cdot \frac{1}{10} + 4 \cdot \frac{1}{2} + 4 \cdot \frac{1}{10} = \frac{2+3+3+4+20+4}{10} = \\ &= \frac{36}{10} = \frac{18}{5} \end{aligned}$$

METODO ALTERNATIVO ULTERIORE

$$E[Y] = E[X_1] + E[X_2].$$

lineare

Si danno anche

$$P_{X_1}(1) = \frac{3}{10}, \quad P_{X_1}(2) = \frac{6}{10}, \quad P_{X_1}(3) = \frac{1}{10}$$

P_{X_2} stessa cosa

$$E[X_1] = E[X_2] = 1 \cdot \frac{3}{10} + 2 \cdot \frac{6}{10} + 3 \cdot \frac{1}{10} = \frac{3 + 12 + 3}{10} = \frac{18}{10}$$

$$E[Y] = \frac{18}{10} + \frac{18}{10} = \frac{36}{10} = \frac{18}{5}.$$